Linear systems and elimination

Optimal uniform row-deletion singular-value constant

Difficulty: challenging **Importance:** interesting to the community

Status: Lean verified

Independently reviewed resolution - 2026-09-11

Affirmative sharp-constant resolution. Theorem 2 and Sections 6-7 prove the sharp constant $c_\theta = \sqrt{h_\theta}$. The upper bound is uniform over all unit-row matrices and even all $m \geq 1$ for sufficiently large n, with squared normalized error $O_\theta(n^{-1/6})$. Theorem 1 supplies matching spherical realizations along every high-aspect-ratio sequence, proving the exact eventual-uniform optimality quantifiers.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Complete manuscript, independent proof review, and submission record. The supplied notes were reconstructed with substantial AI assistance. This is independent agent verification, not external human peer review or formal proof-assistant certification; no novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects an optimal uniform extremum over matrices and deleted row sets; community impact is a sharp robustness limit for row-sampling methods.

Lean proof and verification evidence

Lean verified. The formalization proves the optimal eventual uniform row-deletion constant and shows that every smaller real constant fails. It also gives the squared upper bound for every positive row count and near-sharp unit-row matrix witnesses beyond every dimension and aspect-ratio threshold.

The independent reviews and isolated Linux checks cover the full original target in Lean 4.33.1. The proof uses the standard three axioms. Immutable proof and dependency pins · Reviews, reproduction, and verification logs.

Problem statement

Fix $0 < \theta < 1$. For a real $m \times n$ matrix A with every row of Euclidean norm one, put

$$s_\theta(A) = \min_{S \subseteq \{1,\dots,m\},\ |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2 = 1} \|A_S x\|_2, \qquad M_{m,n}(\theta) = \sup_{\|a_i\|_2 = 1} \sqrt{\frac{n}{m}}\, s_\theta(A).$$

Here a_i is row i of A, and A_S contains exactly the rows indexed by S.

Determine the sharp asymptotic upper constant for $M_{m,n}(\theta)$ as $n \to \infty$ and $m/n \to \infty$. In particular, is it

$$c_\theta = \left(\frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2}\, dt \right)^{1/2}, \qquad \mathbb{P}(|G| \leq a_\theta) = \theta, \quad G \sim N(0,1)?$$

Precisely, the proposed uniform inequality asks whether, for every $\varepsilon > 0$, there exist integers N and R such that

$$M_{m,n}(\theta) \leq c_\theta + \varepsilon \quad \text{whenever } n \geq N \text{ and } m/n \geq R,$$

and whether no smaller constant has this property. The supremum and quantifiers specify the source's uniform $1 + o(1)$ bound; the floor convention handles row-count integrality.

Editorial ratings; a bounded search cannot certify that a problem remains unsolved.
Verification check: 2026-11-10

Connection to numerical linear algebra

This asks how well any unit-row linear system can retain its smallest singular value under worst-case row removal, a geometric constraint on robust iterative solvers.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, the third question immediately after equation (8). Published paper. Author preprint, v1, §2.3, paragraph following (⋄).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1 (background on the same restricted-singular-value heuristic and the different streaming model), §5. Preprint.

Earlier status check — 2026-09-08

Checked the published question and latest arXiv records on 2026-09-08. Targeted searches for "Steinerberger subsingular", "small subsingular values", "quantile Kaczmarz optimal constants", "unit rows smallest singular value", and 2025/2026 variants found no resolution of the all-matrices extremum. Recent streaming Kaczmarz guarantees do not supply this universal extremal inequality. The random-ensemble limit in the same source is a related but distinct question: it specifies a random construction, whereas this question optimizes over all unit-row matrices. Openness remains subject to the limits of this search.

Audit update — 2026-09-10

Rechecked Steinerberger's primary formulation, §2.3. Searches for the optimal row-deletion constant and quantile-Kaczmarz follow-ups found no proof of this extremal value. The new Cai–Chen–Ma–Wu paper, SIAM J. Matrix Anal. Appl. 47 (2026), 802–823, proves a tight QRK subsample-size result, not the universal matrix constant asked for here.